

Gram's Staining Technique

Aim:

To demonstrate the Gram reaction of the given bacterial culture and identify whether it is Gram-positive (or) Gram-negative.

Principle:

Gram's stain was first developed by Hans Christian Gram. It is use full for classifying bacteria into two major groups; gram +ve and gram -ve. In this process the fixed bacterial smear is stained in the order given below;

- Crystal violet (primary stain)
- Gram's iodine (mordant)
- alcohol-acetone (decolorizing agent)
- saffranine (counter stain)

Bacteria which retain the primary stain appear violet or pinkish purple and are termed as gram +ve bacteria, whereas few bacteria lose their primary stain and take up the counter stain and appear pinkish in color and are termed as gram -ve bacteria. The difference in staining response to the gram stain can be attributed to the chemical and the physical difference of their cell wall composition. The gram -ve bacterial cell wall is the complex multilayered structure and contains high lipid content in addition to muco-peptides. The higher amount of lipid is readily dissolved in alcohol resulting in de colorization of bacteria which results in the formation of large pores in the cell wall, resulting in leakage of crystal violet- iodine complex. Because of which the organism take-up the counter stain i.e. saffranine and appear pink. Gram +ve bacteria is rich in cross linked muco- peptides and less amount of lipo-polysaccharide layer. When the gram +ve bacteria is treated with ethanol causes dehydration and reduces the pore size in the cell wall helping in retaining the crystal violet-iodine complex and thereby the cells appear purple. Thus, it cannot take the counter stain saffranine.

- Gram +ve bacteria: The cell wall contains a large amount of peptidoglycan and teichoic acid, which consist of polymers of ribitol phosphate and (or) glycerol phosphate.

Reagents in the order used	Reactions & appearance of bacteria	
	(Gram +ve)	(Gram -ve)
Crystal violet	Cells stain violet	Cells stain violet
Gram's iodine (mordant)	CV-I complex formed within cells. Cells remain violet.	CV-I complex formed within cells. Cells remain violet.
Acetone -alcohol (decolorizer)	Cell walls dehydrate leading to shrinkage of pores, permeability	Lipid is extracted from cells walls. Permeability increases, CV-I complex is removed from cells.

	decreases, CV-I complex cannot be removed from cells.	
Saffranine	Cells not affected and remains violet.	Cells take up saffranine & become pink.

- Gram -ve bacteria: the cell wall contains only a small amount of peptidoglycan. The outer layer of the cell wall however contains an outer membrane made up of toxic lipopolysaccharide molecules, referred to as endotoxin". The lipid 'A' part of the molecules is highly toxic.

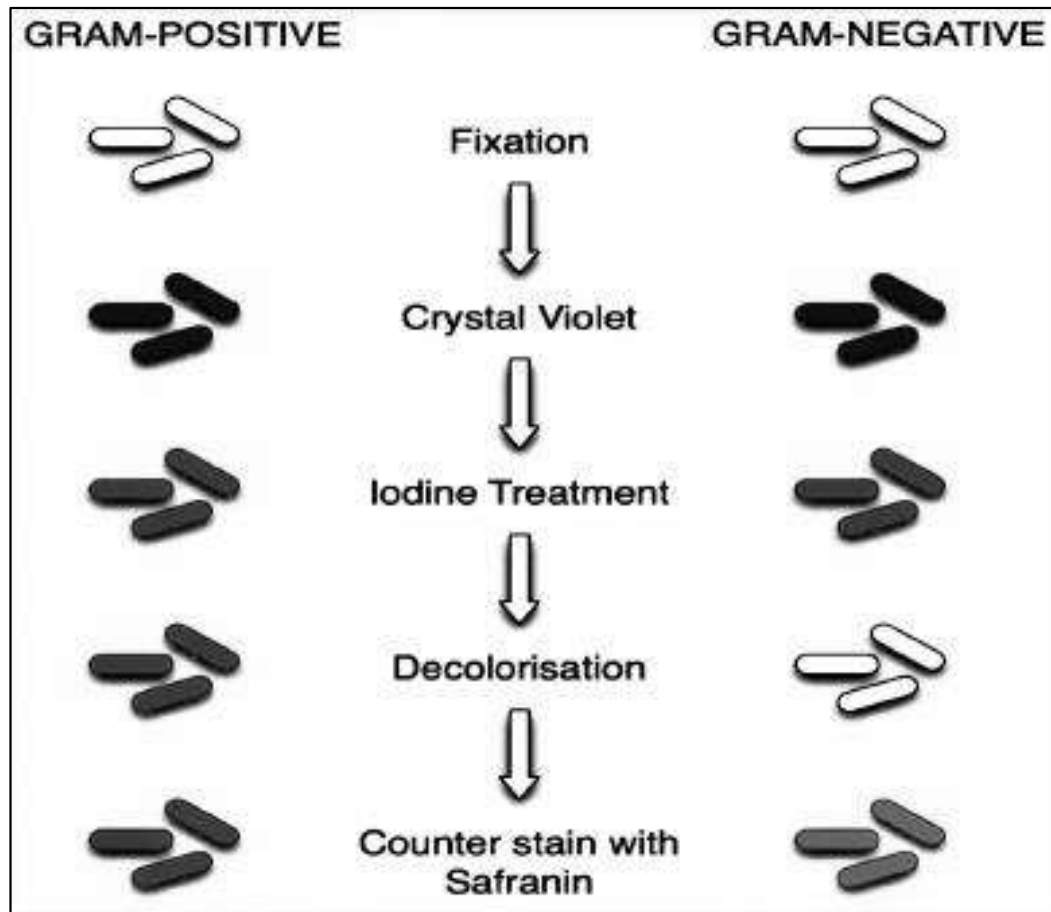
Hence Gram +ve cells appear violet in color and Gram -ve appears pink in color.

Requirements:

Glass slides, 24hrs old cultures of *E. coli* and *Bacillus subtilis*, crystal violet, potassium iodine (or) iodine solution (grams iodine), acetone alcohol (decolorizer) and saffranine, bacteriological loop.

Procedure:

- Prepare the smear on the slide with the bacterial culture.
- Fix the smear by passing the slide over the flame.
- Stain it for one minute with crystal violet solution and wash it under tap water.
- Apply the Gram's iodine solution that acts as a mordant for one minute and wash under tap water.
- Decolorize with alcohol by adding it drop-wise on the tilted slide until all free blue color has been removed for 20-30sec. Wash immediately under tap water.
- Flood the slide with saffranine (counter stain) for one minute and wash it under tap water and air-dry.
- Examine the smear microscopically first with the 40X objective to check the staining and to see the distribution of materials and then with the oil immersion objective to look for bacteria and cells.



Reporting of the grams smear:

The number of bacteria present: many/moderate/few

The grams reaction of bacteria: Gram +ve (purple) or gram -ve (pink)

Morphology of bacteria: Cocci/Bacillus/Coccobacilli

Arrangement of cells: Pairs/Groups/Chains/Singles

Special structures: spores/capsules etc

Proposed Results:

- Gram positive bacteria ----- dark purple to violet
- Gram negative bacteria ----- pale to dark pink or red

Gram-positive bacteria: Staphylococcus, Streptococcus, Corynebacterium, Clostridium, Lactobacillus etc.

Gram-negative bacteria: Neisseria, Haemophilus, Salmonella, Shigella, Vibrio, Klebsiella, Coliforms, Brucella, Yersinia etc.

Variations in gram reaction:

Occasionally gram +ve organisms lose their ability to retain crystal violet and take up the counter stain due to following

- Cell wall damage due to excessive heat fixation of the smear and over decolorization of the smear
- Use of iodine solution which is too old and yellow instead of brown and not acting as an effective mordant
- Preparation of the smear from an old culture, then the gram +ve organism may not be fully decolorized and may appear as gram -ve if the smear is too thick.

Observation:

Culture A:

Staining property: Cells appear pink or pinkish red

Cell morphology: Rod shaped bacteria

Cell arrangement: Cells are arranged single

Inference:

As the cells appear as pink rods, it is concluded that the given culture is of gram -ve with bacilli arranged in singles.

Culture B:

Staining property: Cells appear purple or violet

Cell morphology: Small spherical shaped Cocci

Cell arrangement: Cells are arranged in bundles

Inference:

The given culture is of gram +ve Cocci arranged in bundles.

